(2) m is 1, then the required factor is $k(a + b + c)$ and

(3) m is 2, then the required factor is $k(a^2 + b^2 + c^2) + l\,(ab + bc + ca)$.

Example 7.23

Using Factor Theorem, prove that $\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = (x-1)^2(x+9)$.

Solution

$$\text{Let } |A| = \begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix}.$$

$$\text{Putting } x = 1, \text{ we get } |A| = \begin{vmatrix} 2 & 3 & 5 \\ 2 & 3 & 5 \\ 2 & 3 & 5 \end{vmatrix} = 0$$

Since all the three rows are identical, $(x - 1)^2$ is a factor of $|A|$

$$\text{Putting } x = -9 \text{ in } |A|, \text{we get } |A| = \begin{vmatrix} -8 & 3 & 5 \\ 2 & -7 & 5 \\ 2 & 3 & -5 \end{vmatrix} = \begin{vmatrix} 0 & 3 & 5 \\ 0 & -7 & 5 \\ 0 & 3 & -5 \end{vmatrix} = 0$$

Therefore $(x+9)$ is a factor of $|A|$ [since $C_1 \to C_1 + C_2 + C_3$].

The product $(x - 1)^2 (x + 9)$ is a factor of $|A|$. Now the determinant is a cubic polynomial in x.

Therefore the remaining factor must be a constant 'k'.

$$\text{Therefore } \begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = k(x-1)^2(x+9).$$

Equating x^3 term on both sides, we get $k = 1$. Thus $|A| = (x - 1)^2 (x + 9)$.

Example 7.24

Prove that $\begin{vmatrix} 1 & x^2 & x^3 \\ 1 & y^2 & y^3 \\ 1 & z^2 & z^3 \end{vmatrix} = (x-y)\,(y-z)\,(z-x)\,(xy+yz+zx)$.

Solution

$$\text{Let } |A| = \begin{vmatrix} 1 & x^2 & x^3 \\ 1 & y^2 & y^3 \\ 1 & z^2 & z^3 \end{vmatrix}.$$

$$\text{Putting } x = y \text{ gives } |A| = \begin{vmatrix} 1 & y^2 & y^3 \\ 1 & y^2 & y^3 \\ 1 & z^2 & z^3 \end{vmatrix} = 0 \quad (\text{since } R_1 \equiv R_2).$$

Therefore $(x - y)$ is a factor.

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